# t3-w09 — condition C0

Need for statistic — Need for statistics: purposes, and the collection and tabulation of data

> The curriculum objective alone. No lesson plan, no note.

## INPUT LAYERS

| Input layer | What it carries | Channel | In this condition |
|---|---|---|---|
| Scope | what is to be taught, and to whom | prompt | **present** |
| Prior knowledge | what the learner already knew coming in | prompt | absent |
| Materials | what the lesson was represented with | prompt | absent |
| Position | where this objective sat in the sequence | prompt | absent |
| Method | the shape of the teaching, as event labels | prompt | absent |
| Enacted content | what was actually explained and worked through | prompt | absent |
| Set work | what the learners were asked to do in class | prompt | absent |
| Assessment | the questions the lesson was checked with | prompt | absent |
| Figures | the drawings the lesson used | tool | absent |

## WHAT THE MODEL IS SENT

Contract `ComposeWriteObjectiveSection`, declaring: `artefact_spec, lesson_context, unit_context`.

### Lesson context — shared by every section

**`subject`**

```
Mathematics
```

**`grade_level`**

```
JSS1
```

**`topic`**

```
Need for statistic
```

**`subtopic`**

```
Need for statistics: purposes, and the collection and tabulation of data
```

**`term`**

```
Third Term
```

**`week`**

```
9
```

**`lesson_objectives`**

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

### Objective 1 — explain the purpose of statistic

**`objective_text`**

explain the purpose of statistic

**Withheld from this payload:** `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

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### Objective 2 — analyse and interpret statistical data presented in tables

**`objective_text`**

analyse and interpret statistical data presented in tables

**Withheld from this payload:** `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `current_step`, `instructional_events`, `student_tasks`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state` — carried by the plan or the unit, not sent under this condition.

## WHAT THE MODEL PRODUCED

Objective 1

### Why Statistics Matter: Understanding Its Purpose

EXPLANATION

Statistics is the science of collecting, organising, presenting and interpreting data. Its main purpose is to help us make sense of the world by turning raw numbers into useful information. By summarising large amounts of data, statistics allows us to:

1. **Identify patterns and trends** – for example, seeing how temperatures change over a year.
2. **Make predictions** – such as estimating how many students will need new textbooks next term.
3. **Support decisions** – like choosing the best location for a new school based on population data.
4. **Communicate findings clearly** – using tables, charts and graphs so others can understand the results.

In everyday life, statistics is used in sports (player performance), health (disease rates), business (sales figures) and many other fields. Understanding its purpose helps you see why collecting accurate data and presenting it well is so important.

WORKED_EXAMPLE

**Worked Example:**

A school wants to know the most popular sport among its students. They ask 120 students which sport they prefer and record the answers.

| Sport | Number of Students |
|---|---|
| Football | 45 |
| Basketball | 30 |
| Athletics | 25 |
| Others | 20 |

**Purpose of this table:**
- It **organises** the raw responses into a clear format.
- By looking at the numbers, we can **identify the most popular sport** (football).
- The school can **make a decision** to allocate more resources (e.g., a better football field) based on this information.
- The table also **communicates** the results to teachers, parents and the school board in an easy-to-read way.

**1.** A class surveyed 80 students about their favourite fruit. The results are shown below. | Fruit | Number of Students | |---|---| | Apples | 20 | | Bananas | 15 | | Oranges | 25 | | Others | 20 | What is the purpose of presenting this data in a table? Explain in two sentences.

answer: The table organises the survey results so we can easily see how many students prefer each fruit, and it helps us identify the most popular fruit (oranges) and make decisions, such as which fruit to buy for a class party.

**2.** A local shop recorded the number of customers each day for a week: | Day | Customers | |---|---| | Monday | 45 | | Tuesday | 50 | | Wednesday | 55 | | Thursday | 48 | | Friday | 60 | | Saturday | 80 | | Sunday | 70 | Why is it useful to tabulate this information? Give one reason related to decision-making.

answer: Tabulating the data shows the pattern of customer flow, allowing the shop owner to decide when to schedule more staff, such as increasing staff on Saturday when the number of customers is highest.

**3.** A sports coach recorded the times (in seconds) of 5 runners in a 100-m sprint: | Runner | Time | |---|---| | A | 12.5 | | B | 13.0 | | C | 12.8 | | D | 13.2 | | E | 12.9 | Explain how a table helps the coach understand the runners' performance.

answer: The table arranges each runner’s time in one place, making it easy to compare performances and identify the fastest runner (Runner A) and see overall speed trends.

Objective 2

### Reading and Interpreting Data Tables

> EXPLANATION
> A data table arranges information in rows and columns so you can see patterns quickly. Each column has a heading that tells you what the numbers in that column represent, and each row usually represents a different item or observation. To interpret a table you should:
> 1. Identify what each column measures (e.g., number of students, scores, ages).
> 2. Look for the highest and lowest values.
> 3. Notice any trends – does the value increase, decrease, or stay the same?
> 4. Compare totals or averages if they are given, or calculate them yourself.
> 5. Answer the question asked by using the relevant part of the table.

> WORKED_EXAMPLE
> Example:
>
> | Class | Number of Students |
> |---|---|
> | A | 28 |
> | B | 32 |
> | C | 25 |
> | D | 30 |
>
> **Interpretation:**
> - The class with the most students is B (32 students).
> - The class with the fewest students is C (25 students).
> - Total number of students = 28 + 32 + 25 + 30 = 115.
> - Average number of students per class = 115 ÷ 4 = 28.75.
>
> So, most classes have around 29 students, but Class B is a little larger.

**1.** Use the table below to answer the questions. | Fruit | Quantity | |---|---| | Apples | 12 | | Bananas | 8 | | Oranges | 15 | | Grapes | 5 | a) Which fruit has the highest quantity? b) What is the total quantity of all fruits? c) What is the average quantity per fruit (round to one decimal place)?

answer: a) Oranges b) 40 c) 10.0

**2.** The table shows the number of books read by four students in a month. | Student | Books Read | |---|---| | Lina | 7 | | Tunde | 5 | | Aisha | 9 | | Emeka | 6 | Answer the following: 1) Who read the most books? 2) What is the total number of books read? 3) If the goal is at least 8 books, how many students met the goal?

answer: 1) Aisha 2) 27 3) 1

**3.** A survey recorded the number of pets owned by families in a neighbourhood. | Number of Pets | Families | |---|---| | 0 | 4 | | 1 | 7 | | 2 | 5 | | 3 | 2 | Find: - The total number of families surveyed. - The total number of pets owned. - The average number of pets per family (to two decimal places).

answer: Total families: 18 Total pets: 23 Average pets per family: 1.28

## LAYER BY LAYER: WHAT WAS GROUNDED AND WHAT WAS NOT

| Output layer | Made | Written from | Verdict |
|---|---|---|---|
| Title<br>the reader-facing framing of the section | 2 | Scope | **grounded** |
| Explanation<br>the principle, rule or relationship to review | 2 | nothing in the input | **unsourced** |
| Worked example<br>a problem with its solution worked through | 2 | nothing in the input | **unsourced** |
| Figure<br>a drawing the learner reads from | 0 | nothing in the input | not produced |

| | | |
|---|---|---|
| Practice<br>what the learner does | 6 nothing in the input | **unsourced** |
| Answer<br>what the learner checks against | 6 nothing in the input | **unsourced** |